

Name	Index Number	Form Class	Tutorial Class	Subject Tutor
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ANGLO-CHINESE JUNIOR COLLEGE
DEPARTMENT OF CHEMISTRY
Promotional Examination

CHEMISTRY
Higher 2

9729

Answer Booklet

1 October 2018

2 hours

Additional Materials: Question Booklet
 Data Booklet
 Optical Answer Sheet (OAS)
 Writing Papers

READ THESE INSTRUCTIONS CAREFULLY

Write your name and class on all the work you hand in. Write in dark blue or black pen. You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluids.

You are reminded of the need for good English and clear presentation in your answers. The number of marks is given in brackets [] at the end of each question or part question.

Shade your answers to Section A in the Optical Answer Sheet (OAS) provided. **No extra time will be given for the shading of answers.**

Answer **all** questions of Section B in the Answer Booklet.

Answer **all** questions of Section C on writing paper. Fasten your answers to Section C to this booklet.

For Examiner's Use		
	Question No.	Marks
Section A (Total: 15 m)		
Section B (Total: 35 m)	1	
	2	
	3	
Section C (Total: 25 m)	1	
	2	
	3	
	4	
TOTAL: 75 m		

This document consists of **9** printed pages, including this cover page.



Section B (35 marks)

Answer **all** questions in the spaces provided.

1 Uranium, U, is an important element in nuclear research.

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(a) Uranium-235 (^{235}U) is an isotope of uranium making up less than one percent of natural uranium.

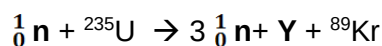
(i) State the number of neutrons and protons in a ^{235}U atom. [1]

(ii) All the isotopes of uranium are unstable. [1]

When a ^{235}U atom is bombarded with a neutron, three neutrons, an atom of krypton-89 and a particle, **Y** are formed.

Y contains 88 neutrons and other particles.

Schematically,



Determine the identity of **Y**.

(b) The most stable isotope of uranium is uranium-238.

For **(b)**, you are to use 238.0 as the relative atomic mass of uranium.

Excess water was added to 4.267 g of UF_6 .

The only products were 3.730 g of a solid **Z** containing only uranium, oxygen and fluorine and 0.970 g of HF gas. The molecular formula of **Z** contains one U atom.

(i) Calculate the number of moles of HF and UF_6 . [2]

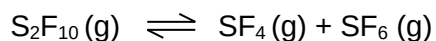
- (ii) Hence, state the number of moles of **Z** formed. [1]
- (iii) Using your answer in **b(ii)**, determine the molar mass of **Z**. [1]
- (iv) Using your answers in **(b)**, determine the molecular formula of **Z**. [3]

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[Total: 9]

- 2 Disulfur decafluoride (S_2F_{10}) is an extremely toxic gas which almost became a chemical weapon during World War II.

It exists in equilibrium with sulfur tetrafluoride (SF_4) and sulfur hexafluoride (SF_6).



- (a) There is a S-S single bond in disulfur decafluoride. All the ten fluorine atoms are in equivalent environments in disulfur decafluoride.

- (i) State the shape with respect to each sulfur in S_2F_{10} . [1]

- (ii) The boiling points of the three compounds are tabulated below.

[2]

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Compound	SF ₄	SF ₆	S ₂ F ₁₀
Boiling point / K	235	209	303

Explain why SF₆ has a lower boiling point than SF₄.

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- (iii) Explain why SF₄ has a lower boiling point than S₂F₁₀.

[1]

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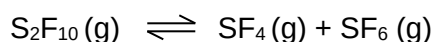
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- (b) This part of the question focuses on the thermodynamic aspects of the reaction.



- (i) Using suitable data from the Data Booklet, determine the enthalpy change of the reaction. [2]

State the most important assumption you made.

- (ii) Explain why this decomposition becomes more thermodynamically feasible at higher temperatures. [2]

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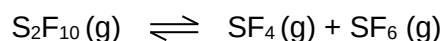
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- (c) This part of the question focuses on the chemical equilibrium aspects of the reaction.



In a study of this decomposition, 2 moles of disulfur decafluoride was placed in a 2.00 dm³ flask and heated up to 100 °C; its concentration became 0.50 mol dm⁻³ at equilibrium.

- (i) Calculate the value of K_c at 100 °C. Include the units. [2]

To this equilibrium mixture, a certain mass of disulfur decafluoride was added. The new equilibrium concentration of it became 2.50 mol dm^{-3} .

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- (ii) Calculate the mass of disulfur decafluoride added. [3]

- (iii) Predict, with reasoning, how the *position of equilibrium* would change when the volume of the container is reduced. [1]

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[Total: 14]

- 3 This question is about interhalogens, compounds that are made of more than one type of halogen.

Chlorine trifluoride, ClF_3 , is one of the most reactive chemicals ever known to man. It causes sand to explode.

- (a) ClF_3 reacts with silver chloride according to the **unbalanced** equation below.



- (i) Balance the above equation by inserting the respective stoichiometric coefficients in the spaces provided below. Do fill in the coefficient even if it happens to be 1. [1]



- (ii) State the electronic configuration of the outermost shell of Ag in AgCl . [1]

- (b) ClF_3 and ClBr_3 share the same molecular shape. However, the F-Cl-F bond angle is predicted to be smaller than the Br-Cl-Br angle.

- (i) Explain why bromine is less electronegative than fluorine. [1]

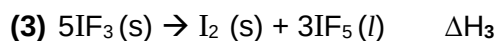
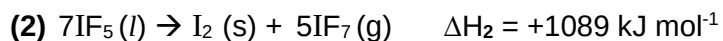
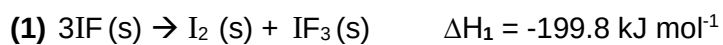
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- (ii) Hence, or otherwise, explain the relative sizes of the two bond angles. [2]

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- (c) Iodine forms many compounds with fluorine – IF, IF₃, IF₅ and IF₇.

The four compounds can interconvert into one another according to the following chemical equations.



- (i) All three reactions can be classified under a certain type of redox reaction. Identify this type of reaction. [1]

The standard enthalpy changes of formation of IF and IF₇ are -95.4 and -962.5 kJ mol⁻¹ respectively.

- (ii) Calculate the standard enthalpy changes of formation of IF₃ and that of IF₅. [2]

- (iii) Hence, calculate the standard enthalpy change of reaction (3). [1]

- (iv) Briefly explain why there are negligible changes in entropy in reactions **(1)** and **(3)**. [1]

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- (v) Hence, predict whether reaction **(1)** or **(3)** has a lower likelihood of happening. [2]

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[Total: 12]

~ END OF SECTION B ~